



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware(✓)/ Software Report-1 Submission

Semester: Spring 26

Course Code: EEE283L

Course Title: Digital Logic Design

Section:03

Experiment No:01

Experiment Name: Familiarization of Fundamental Logic Gates

Group No: 02

Group members:

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Date of Submission: 01.03.2026

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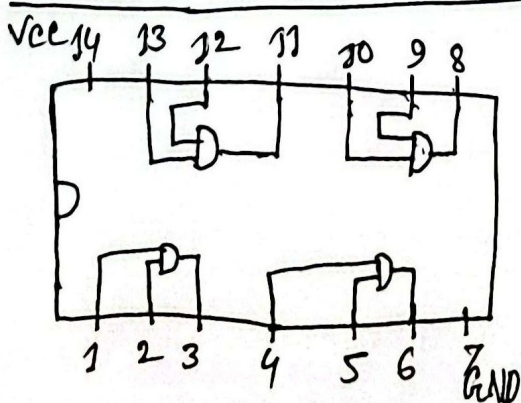
Objective:

The objective of this experiment are to understand the operation of basic and universal logic gates also get Practical experience with circuit implementation and. construct simple logic circuit using these gates.

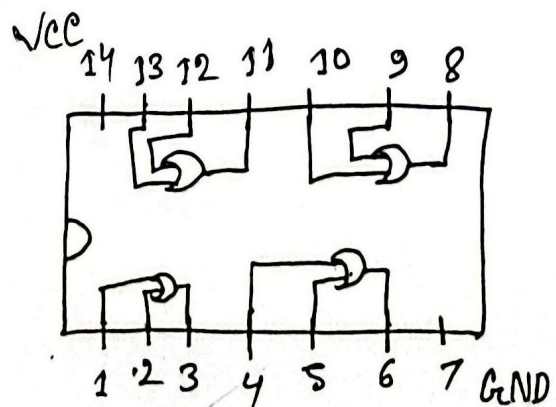
Equipment

- 1) Digital trainer board.
- 2) DC Power supply
- 3) Jumper wires
- 4) Logic gate ICs:
 - AND - 7408
 - OR - 7432
 - NOT - 7404
 - NAND - 7400

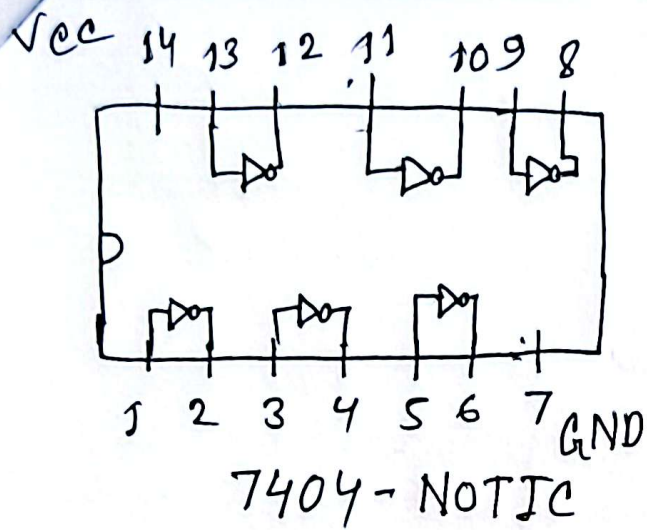
Logic gate IC's structure



7408 - AND IC



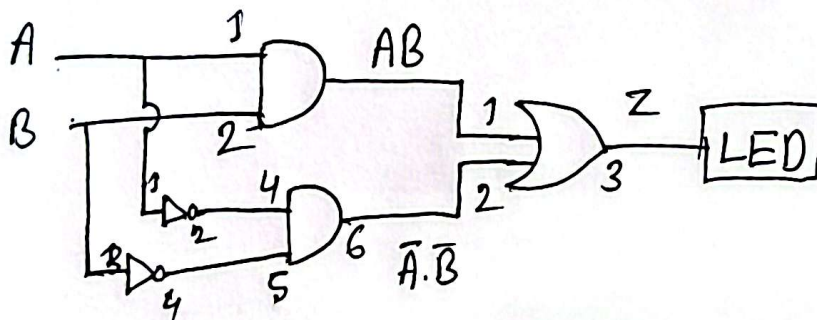
7432 - OR IC



Experimental setup

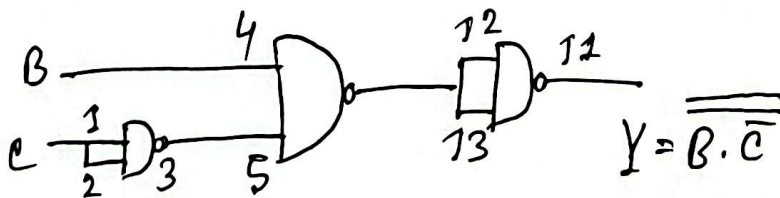
Part-A

Using ~~only~~ fundamental gates only



Part-B

Using NAND gates only



Simulation Diagram:

Using Fundamental gates only

Part:A

Equation:- $Z = A.B + \bar{A}. \bar{B}$

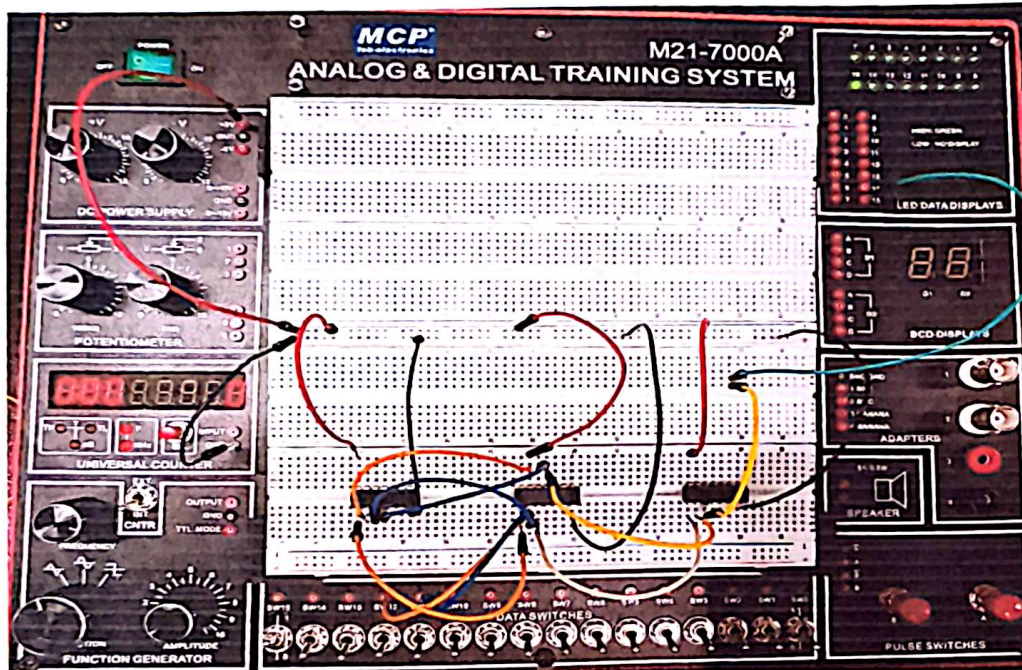


Figure-1: Circuit behavior for On condition [input(A=0,B=0)], (Output: High)

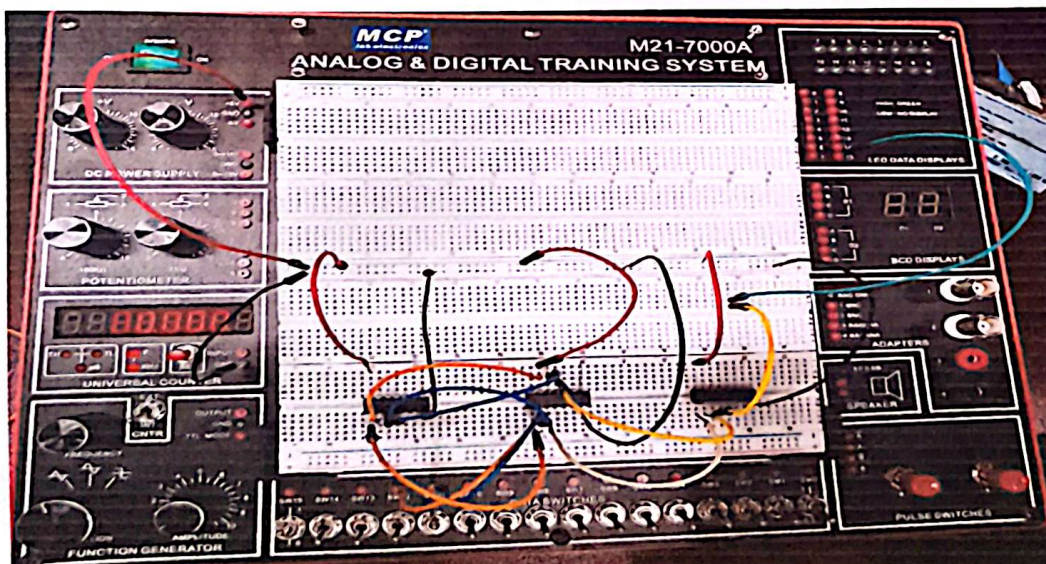


Figure-2: Circuit behavior for Off condition [input(A=0,B=1)], (Output: ~~High~~
Low)

Part:B

Using NAND gates only

Equation:- $Z = B \cdot \bar{C}$

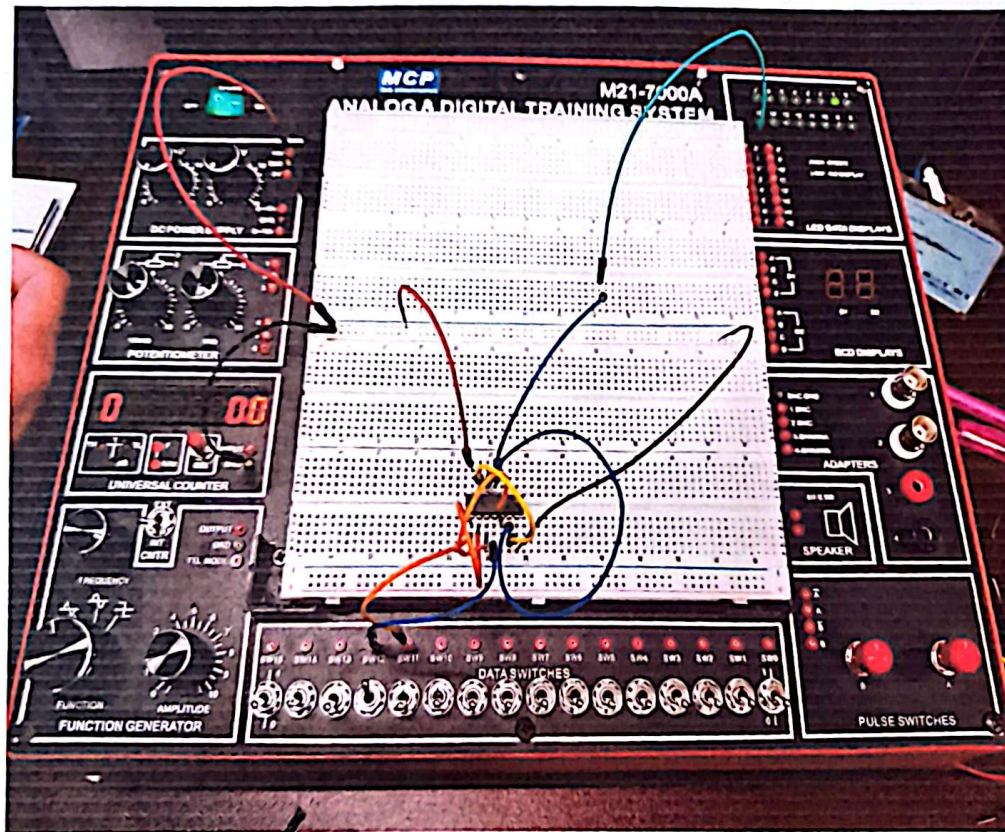


Figure-3: Circuit behavior for On condition [input(B=1,C=0)], (Output: High)

Results

Truth Tables.

Part-A, Equation:- $Z = A \cdot B + \bar{A} \cdot \bar{B}$

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	1

Part-B

Equation:- $Y = \overline{B \cdot \bar{C}} = B \cdot \bar{C}$

B	$\bar{C}$	Y
0	0	0
0	1	0
1	0	1
1	1	0

Questions

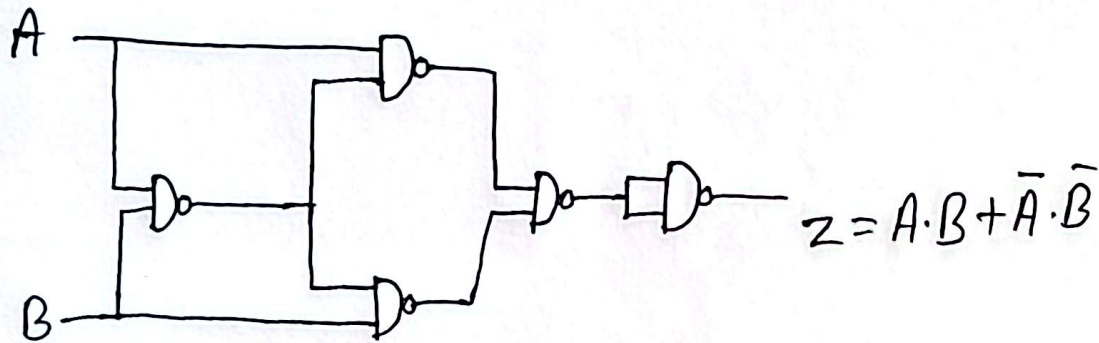
Questions

ii) NAND and NOR gates are called 'universal gates' ~~because~~ because any other logic gate can be built using ~~only~~ only NAND or only NOR gates. Also, They are capable of performing all basic logical operation.

Here in the NAND gate representation of the experiment circuit.

Part-A

$$Z = A \cdot B + \bar{A} \cdot \bar{B}$$



iii) Here, $Y = (\bar{A} + B) \cdot (A + B + C) \cdot \bar{C}$

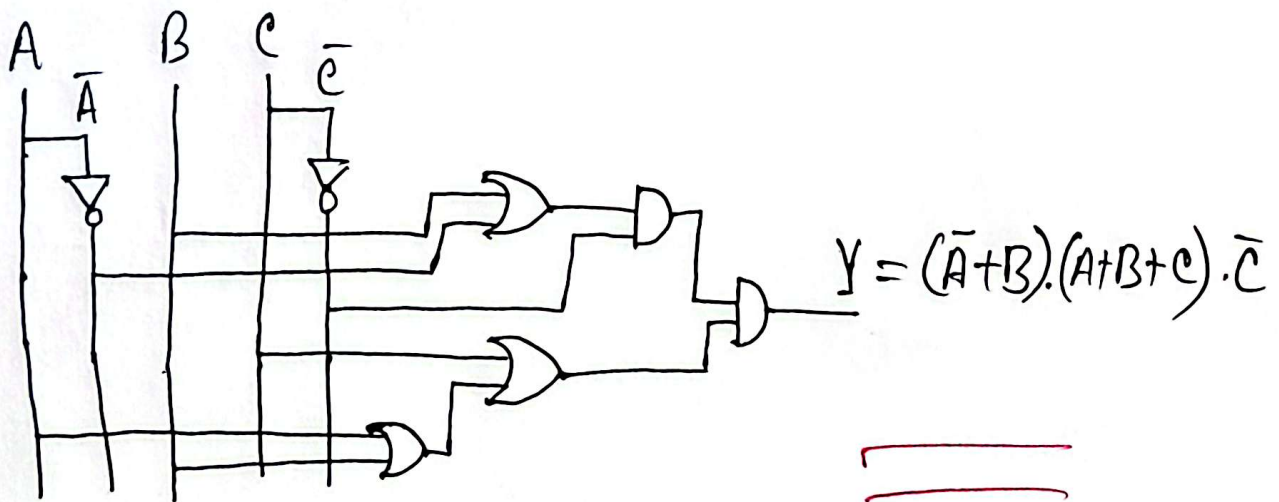


Figure :- Logic Design based on Fundamental gates.

~~He~~

Here,

$$\begin{aligned}
 Y &= (\bar{A}+B) \cdot (A+B+C) \cdot \bar{C} \\
 &= [\bar{A}(A+B+C) + B(A+B+C)] \cdot \bar{C} \\
 &= [\bar{A}A + \bar{A}B + \bar{A}C + BA + BB + BC] \cdot \bar{C} \\
 &= [0 + \bar{A}B + \bar{A}C + AB + B + BC] \bar{C} \\
 &= [B + \bar{A}C + B + BC] \bar{C} \quad [\bar{A}B + AB = B] \\
 &= [B + \bar{A}C + BC] \bar{C} \\
 &= [B(1+C) + \bar{A}C] \bar{C} \\
 &= (B + \bar{A}C) \cdot \bar{C} \\
 &= B\bar{C} + \bar{A}C\bar{C} \\
 &= B\bar{C}
 \end{aligned}$$

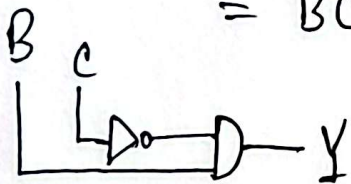


Figure: Logic design by using only fundamental gates,

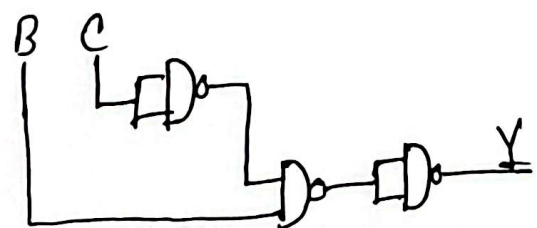


Figure: Logic design by using only NAND gates.

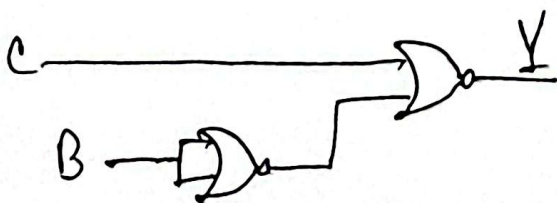


Figure: Logic design by using only NOR gates.

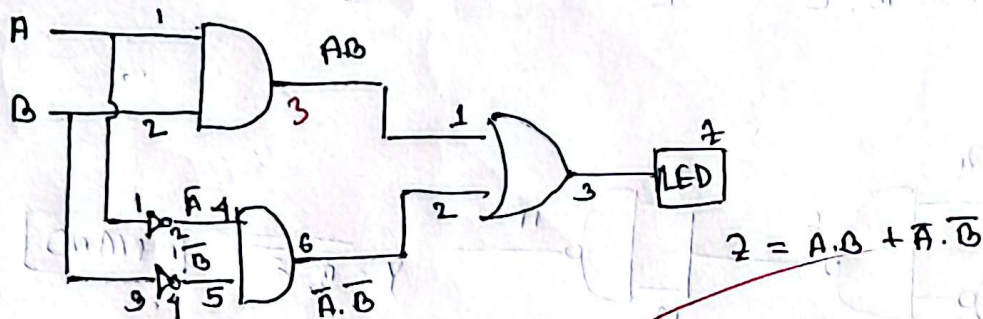
Discussion

In this experiment, we explored how digital logic works by building circuits using both fundamental and universal gates. By constructing the circuit for $z = A \cdot B + \bar{A} \cdot \bar{B}$ and $z = \bar{B} \cdot \bar{C}$, we saw that how basic gates like AND, OR, NOT works. Also we see the universal gate representation. Here, we simplifying $y = (\bar{A} + B) \cdot (A + B + C) \cdot \bar{C}$ down to $B\bar{C}$ and rebuild it by using NAND OR NOR gates. It proves that why it called universal gate. we learned that right output ~~come~~ depends on both smart math and careful wiring, as even a small mistake on the trainer board can change the entire result.



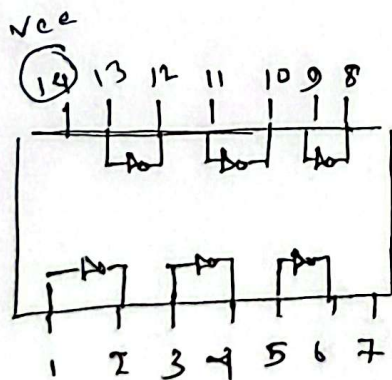
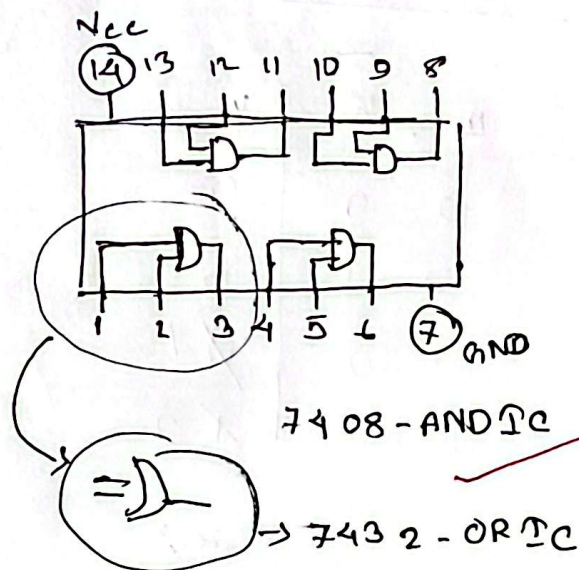
Appendix:-

Exp:1 Familiarization & construction of logic gates



Truth Table

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	1

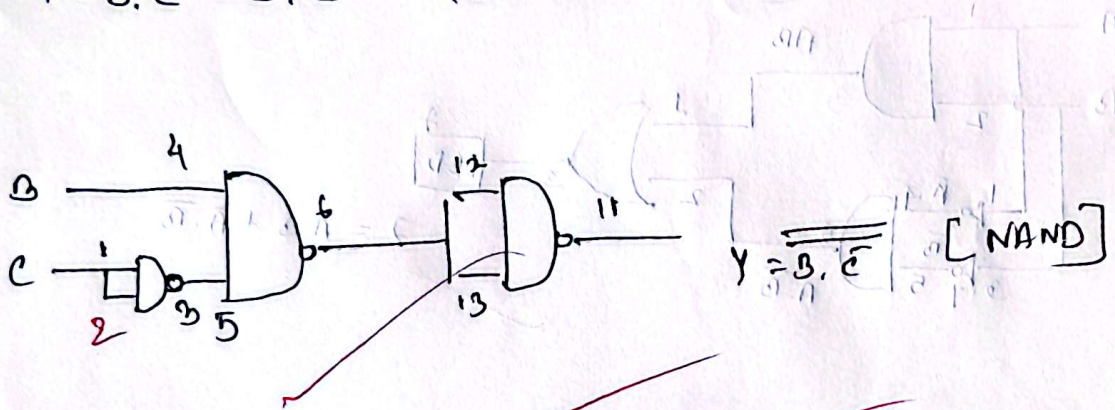


A.B	AND	OR	NAND	NOR
00	0	0	1	1
01	0	1	1	0
10	0	1	1	0
11	1	1	0	0

Part 1: B

Universal Gate:- NAND, NOR

$$Y = B \cdot \bar{C} = \overline{\overline{B \cdot \bar{C}}} \quad [\text{Boolean } \overline{\overline{A}} = A]$$



Truth Table:-

B	$\bar{C}$	Y
0	0	0
0	1	0
1	0	0
1	1	1

22/2/21

B	$\bar{C}$	Y
0	0	0
0	1	0
1	0	0
1	1	1

B	$\bar{C}$	Y
0	0	0
0	1	0
1	0	0
1	1	1